

Nuclear Astrophysics (NS 3035)

Lec 4: Thermonuclear Reactions I

ITER tokamak. The tokamak is a donut-shaped magnetic confinement device designed to sustain extremely hot plasma for nuclear fusion experiments.

Dehiwalage Don Dilruwan

(Photo credit: ITER)

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Core Nuclear Physics Concepts

Reaction Probabilities and Reaction Rates

Neutron-Induced Reactions

Thermonuclear Reactions in Stars

We are made of star stuff — Carl Sagan

Every atom of carbon, oxygen, calcium, and iron in our bodies was forged through nuclear reactions inside stars.

Why Study Thermonuclear Reactions?

Fundamental Questions

- I. Why do stars shine for billions of years?
- II. How do nuclei overcome Coulomb repulsion and fuse?
- III. What determines the lifetime and evolution of stars?
- IV. How are the chemical elements synthesized in the universe?
- V. How do nuclear reactions shape supernovae, neutron stars, and galaxies?

What We Will Cover Over the Next Few Weeks

Core Nuclear Physics Concepts

- I. Nuclear binding energy, mass defect, and Q-values
- II. Nuclear cross sections and stellar reaction rates
- III. Resonant and non-resonant reactions
- IV. Inverse reactions and detailed balance

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Astrophysical Reaction Mechanisms

- V. Neutron-induced reactions
- VI. Gamma-induced reactions
- VII. Charged-particle induced reactions
- VIII. Charged-particle non-resonant reactions
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Outline

Thermonuclear Reactions in Stars

Core Nuclear Physics Concepts

Reaction Probabilities and Reaction Rates

Neutron-Induced Reactions

Part I

Core Nuclear Physics Concepts

Basic Nuclear Physics Concepts and Nomenclature

Atomic Structure

Atom = nucleus + electrons

Nucleus contains **protons** (p , charge $+e$) and **neutrons** (n , charge 0)

Nuclei are denoted as A_ZX

Z = atomic number = number of protons

N = neutron number

$A = Z + N$ = mass number (${}^{12}_6\text{C}$)

Isotopes: same Z , different N

Nuclide: specific nucleus with fixed Z, N

Nuclear Binding

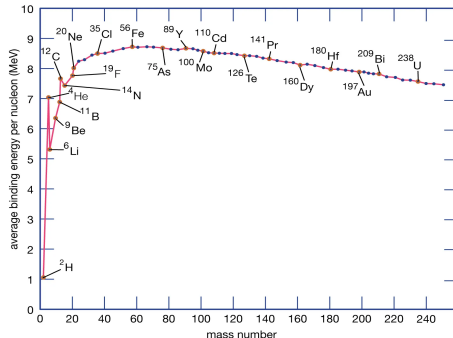
Binding Energy is the energy released when free nucleons combine to form a bound nucleus

Nuclear mass defect $\Delta m = Zm_p + Nm_n - m_{\text{nucleus}}$

$B = \Delta mc^2$ (B = binding energy)

Larger $B/A \Rightarrow$ more stable nucleus

B/A rises rapidly for light nuclei



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A Conceptual Challenge

Yesterday, I gave you a small challenge related to nuclear fusion.

The Question

Imagine that two light nuclei fuse to form a heavier nucleus. Experimentally, we find that the mass of the final nucleus is smaller than the combined masses of the original nuclei.

$$m_{\text{final}} < m_1 + m_2$$

So where did the “missing mass” go?

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$$m_{\text{final}} < m_1 + m_2$$

So where did the “missing mass” go?

Was mass destroyed?

Is mass not conserved?

Or is mass actually another form of energy?

Where Did the Missing Mass Go?

The Key Idea

The “missing mass” was not destroyed. It was converted into released energy according to Einstein's relation $E = mc^2$

Physical Picture

Initially, two separate nuclei have a certain total energy. As they approach each other, the **strong nuclear force** becomes attractive. The nucleons fall into a deep nuclear potential well and become more **tightly bound**. The final bound nucleus therefore has lower total energy.

“Lower total energy means lower total mass”

Mass Defect and Binding Energy

$$\Delta m = Zm_p + Nm_n - m_{\text{nucleus}}$$

$$B = \Delta mc^2$$

Δm : mass defect

B : nuclear binding energy

Where Did the Missing Mass Go?

The Key Idea

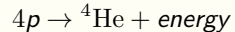
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“Lower total energy means lower total mass”

Fusion Example



The helium nucleus is more tightly bound than four isolated protons.

The missing mass appears as:

kinetic energy

gamma rays

The Three Fundamental Interactions in Nuclear Astrophysics

Strong Interaction

The strong interaction binds protons and neutrons together inside atomic nuclei. It is the strongest known fundamental force, but it acts only over very short distances of about

$$\sim 1 \text{ fm}$$

At these short distances, the strong force overcomes the electromagnetic repulsion between positively charged protons. This interaction is responsible for nuclear binding, nuclear structure, and fusion reactions inside stars.

Astrophysical Importance:

The competition between the strong and electromagnetic interactions determines whether fusion can occur, while weak interactions control stellar evolution timescales and neutrino production.

The Three Fundamental Interactions in Nuclear Astrophysics

Electromagnetic Interaction

The electromagnetic interaction acts between electrically charged particles and has an infinite range. In nuclei and stars, it produces the Coulomb repulsion between positively charged nuclei:

$$V_C(r) = \frac{Z_1 Z_2 e^2}{r}$$

This Coulomb barrier makes nuclear fusion difficult because nuclei naturally repel each other. Electromagnetic interactions are also responsible for atomic structure, photon emission, and gamma-ray decay.

Astrophysical Importance:

The competition between the strong and electromagnetic interactions determines whether fusion can occur, while weak interactions control stellar evolution timescales and neutrino production.

The Three Fundamental Interactions in Nuclear Astrophysics

Weak Interaction

The weak interaction is responsible for changing one type of particle into another, especially neutron-proton conversion:

$$n \leftrightarrow p$$

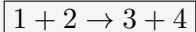
Although much weaker than the strong force, weak interactions are essential in astrophysics. They govern beta decay, neutrino production, neutrino interactions, and the proton-proton chain that powers stars like the Sun.

Astrophysical Importance:

The competition between the strong and electromagnetic interactions determines whether fusion can occur, while weak interactions control stellar evolution timescales and neutrino production.

Nuclear Reaction Q-values

Consider a reaction between two nuclei/particles:



Initial mass: $m_i = m_1 + m_2$

Final mass: $m_f = m_3 + m_4$

The Q-value measures the net energy released or absorbed:

$$Q = (m_i - m_f)c^2$$

$$Q = (m_1 + m_2 - m_3 - m_4)c^2$$

Exothermic and Endothermic Nuclear Reactions

Exothermic Reactions ($Q > 0$)

$$Q = (m_i - m_f)c^2 > 0$$

- a. $m_f < m_i$
- b. Δm is converted into energy.
- c. Reaction releases kinetic energy and radiation:
 $K_f = K_i + Q$
- d. **Example: Proton-Proton Chain:** $p + p \rightarrow d + e^+ + \nu_e$ ($Q \approx 1.44$ MeV)

Endothermic Reactions ($Q < 0$)

$$Q = (m_i - m_f)c^2 < 0$$

- a. $m_f > m_i$
- b. Additional energy must be supplied.
- c. Final kinetic energy decreases: $K_f = K_i + Q < K_i$
- d. **Example: Photodisintegration:** $^{56}\text{Fe} + \gamma \rightarrow 13\alpha + 4n$ ($Q < 0$)

$Q > 0 \Rightarrow$ energy released

$Q < 0 \Rightarrow$ energy absorbed

The sign of the Q-value determines whether a nuclear reaction can act as an energy source or requires energy input from the stellar environment.

Example: Q-value from Nuclear and Atomic Masses

Reaction Example: Proton Capture

$$^{12}\text{C} + p \rightarrow ^{13}\text{N} + \gamma \qquad Q = (m_i - m_f)c^2$$

Method 1: Using Nuclear Masses (Bare Nuclei)

$$Q_{\text{nuc}} = [m(^{12}\text{C})_{\text{nuc}} + m_p - m(^{13}\text{N})_{\text{nuc}}] c^2$$

Masses: $m(^{12}\text{C}) = 11.996709 \text{ u}$, $m_p = 1.007276 \text{ u}$, $m(^{13}\text{N}) = 13.001899 \text{ u}$

$$\Delta m = 11.996709 + 1.007276 - 13.001899 = 0.002086 \text{ u}$$

$$Q_{\text{nuc}} = 0.002086 \times 931.5 \text{ MeV/u}$$

$$Q_{\text{nuc}} \approx 1.94 \text{ MeV}$$

Example: Q-value from Nuclear and Atomic Masses

Reaction Example: Proton Capture

$$^{12}\text{C} + p \rightarrow ^{13}\text{N} + \gamma \qquad Q = (m_i - m_f)c^2$$

Method 2: Using Atomic Masses (Includes Electrons)

Electrons balance automatically ($6 + 1 = 7$), so we can use neutral atom masses:

$$Q_{\text{atom}} = [M(^{12}\text{C}) + M(^1\text{H}) - M(^{13}\text{N})] c^2$$

Masses: $M(^{12}\text{C}) = 12.000000 \text{ u}$, $M(^1\text{H}) = 1.007825 \text{ u}$, $M(^{13}\text{N}) = 13.005739 \text{ u}$

$$\Delta M = 12.000000 + 1.007825 - 13.005739 = 0.002086 \text{ u}$$

$$Q_{\text{atom}} \approx 1.94 \text{ MeV}$$

Using Atomic Masses in Q-value Calculations

Ordinary Nuclear Reactions

For most reactions, we can calculate directly using atomic masses:

$$Q = \left[\sum M_{\text{atom},i} - \sum M_{\text{atom},f} \right] c^2$$

Electron masses cancel automatically.

No need to convert to nuclear masses.

Convenient for fusion, capture, and photodisintegration.

Since $B_e \sim \text{eV--keV} \ll \text{MeV}$, atomic electron binding-energy corrections are neglected.

Rule of Thumb

For ordinary nuclear astrophysics reactions (fusion, capture, photodisintegration, etc.):

Exceptions

Special care is required for:

β^- decay

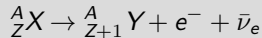
β^+ decay

Electron Capture (EC)

because electron bookkeeping differs between the initial and final atoms.

Q-values for $\beta^\pm$ Decay and Electron Capture

β^- Decay



Using atomic masses:

$$Q_{\beta^-} = [M(X) - M(Y)]c^2$$

Bookkeeping: The emitted electron mass is already accounted for by the addition of an orbital electron in the product atom ($Z+1$).

Important

The need for special formulas arises entirely from **electron-mass bookkeeping**, not from atomic binding energy. $B_e \ll \text{MeV}$ remains true for almost all nuclear astrophysics applications.

Q-values for $\beta^\pm$ Decay and Electron Capture

β^+ Decay



Using atomic masses:

$$Q_{\beta^+} = [M(X) - M(Y) - 2m_e]c^2$$

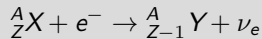
Bookkeeping: The daughter atom has one fewer orbital electron ($Z - 1$), leaving a free electron mass behind. Combined with the emitted positron mass (e^+), a total threshold penalty of $2m_e c^2 \approx 1.022$ MeV must be subtracted.

Important

The need for special formulas arises entirely from **electron-mass bookkeeping**, not from atomic binding energy. $B_e \ll \text{MeV}$ remains true for almost all nuclear astrophysics applications.

Q-values for $\beta^\pm$ Decay and Electron Capture

Electron Capture (EC)



Using atomic masses:

$$Q_{\text{EC}} = [M(X) - M(Y)]c^2$$

Bookkeeping: An inner-shell orbital electron is consumed by the parent nucleus, directly reducing the electron cloud count to match the daughter atom ($Z - 1$). No additional mass correction terms appear.

Important

The need for special formulas arises entirely from **electron-mass bookkeeping**, not from atomic binding energy. $B_e \ll \text{MeV}$ remains true for almost all nuclear astrophysics applications.

Q-value is Not Enough

$Q \longrightarrow$ How much energy?

$\sigma(E), \langle \sigma v \rangle \longrightarrow$ How fast do stars burn?

Nuclear Astrophysics is primarily the study of
reaction probabilities and reaction rates.

Outline

Thermonuclear Reactions in Stars

Core Nuclear Physics Concepts

Reaction Probabilities and Reaction Rates

Neutron-Induced Reactions

Part II

Cross Section and Reaction Rates

Nuclear Reaction Cross Section

What is a Cross Section?

The cross section measures the probability that an incident particle interacts with a target nucleus:

$$\sigma = \frac{\text{Number of reactions per second}}{\text{Incident flux} \times \text{Number of target nuclei}}$$

Classical Interpretation

Treat the nucleus as a small spherical target of radius R . A reaction occurs whenever the projectile hits the target.

$$\sigma = \pi R^2$$

The cross section is simply the static geometric area presented by the nucleus.

Example & Units

For a nucleus with $R \approx 5 \times 10^{-15} \text{ m}$:

$$\sigma = \pi R^2 \approx 8 \times 10^{-29} \text{ m}^2$$

Nuclear physicists use the unit **barn**:

$$1 \text{ barn} = 10^{-28} \text{ m}^2$$

typical cross sections are $\sim$ tenths of a barn.

But...

Crucial Limitation

The classical picture completely ignores the wave nature of particles and cannot explain tunneling through the electrostatic Coulomb barrier.

Nuclear Reaction Cross Section

Wave Mechanics ($\sigma \neq \pi R^2$)

Projectiles act as waves ($\lambda = h/p$). Reaction probability depends on the wave overlap with the nuclear potential. σ depends on:

- I) Energy E & Angular momentum
- II) Nuclear structure selection rules

Coulomb Barrier Tunneling

Classically, if $E < V_C$, no reaction can occur. Quantum mechanically, the tunneling probability is non-zero:

$$P_{\text{tunnel}} > 0$$

This barrier penetration factor strongly suppresses low-energy stellar reaction rates.

Key Result for Stars

The cross section becomes dynamically energy-dependent:

$$\sigma = \sigma(E)$$

It drops exponentially at low energies due to the Coulomb barrier.

Cross Sections and Fundamental Interactions

Geometric Cross Sections (Typical Scales)

For simple geometric scattering (using $1 \text{ barn} = 10^{-24} \text{ cm}^2 = 10^{-28} \text{ m}^2$):

$$^1\text{H} + ^1\text{H} \longrightarrow \sigma = 0.2 \times 10^{-24} \text{ cm}^2 = 0.2 \text{ b}$$

$$^1\text{H} + ^{238}\text{U} \longrightarrow \sigma = 2.8 \times 10^{-24} \text{ cm}^2 = 2.8 \text{ b}$$

$$^{238}\text{U} + ^{238}\text{U} \longrightarrow \sigma = 4.8 \times 10^{-24} \text{ cm}^2 = 4.8 \text{ b}$$

Quantum mechanically, $\sigma = \pi\lambda^2$, where λ is the reduced de Broglie wavelength.

Interaction Dependency at Constant Energy ($E_{\text{proj}} = 2 \text{ MeV}$)

Case 1: The Strong Force

$$^{15}\text{N}(p, \alpha)^{12}\text{C} \longrightarrow \sigma \sim 0.5 \text{ b}$$

- Driven by the **Strong Nuclear Interaction**.
- Highly probable reaction type when the Coulomb barrier is overcome.

Cross Sections and Fundamental Interactions

Geometric Cross Sections (Typical Scales)

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Interaction Dependency at Constant Energy ($E_{\text{proj}} = 2 \text{ MeV}$)

Case 2: The Electromagnetic (EM) Force

$$^3\text{He}(\alpha, \gamma)^7\text{Be} \longrightarrow \sigma \sim 10^{-6} \text{ b}$$

- Driven by the **Electromagnetic Interaction** (gamma emission).
- Suppressed by several orders of magnitude compared to strong interactions.

Cross Sections and Fundamental Interactions

Geometric Cross Sections (Typical Scales)

For simple geometric scattering (using $1 \text{ barn} = 10^{-24} \text{ cm}^2 = 10^{-28} \text{ m}^2$):

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Quantum mechanically, $\sigma = \pi\lambda^2$, where λ is the reduced de Broglie wavelength.

Interaction Dependency at Constant Energy ($E_{\text{proj}} = 2 \text{ MeV}$)

Case 3: The Weak Force

$$p(p, e^+ \nu_e) d \longrightarrow \sigma \sim 10^{-20} \text{ b}$$

- Driven by the **Weak Nuclear Interaction** (beta plus decay during fusion).
- Extremely small cross section; yet to be directly measured experimentally!

Derivation of the Nuclear Cross Section

An incident particle beam hitting a target area A :

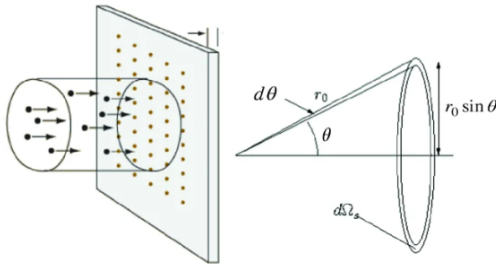


Figure: σ relates incident projectile flux to reaction events, quantifying interaction probability.

- ☐ N_b/t : Incident beam particles per unit time
- ☐ N_t/A : Non-overlapping target nuclei per unit area
- ☐ N_R/t : Reactions occurring per unit time

Derivation of the Nuclear Cross Section

The Cross Section Formula

The cross section σ is defined as the reaction rate per unit incident flux per target particle:

$$\sigma = \frac{N_R/t}{\left(\frac{N_b}{t}\right) \cdot \left(\frac{N_t}{A}\right)} = \frac{\# \text{ of interactions per unit time}}{(\# \text{ of incident particles per unit time}) \times \left(\frac{\# \text{ of target nuclei}}{\text{beam area}}\right)}$$

Units: $[\sigma] \sim \text{cm}^2 \quad (\text{or Barns})$

**Right side of schematic: Cross sections can be integrated over all angles (inclusive) or expressed as differential cross sections ($d\sigma/d\Omega$) based on the scattering solid angle.*

Cross Section for Stellar Environments

In laboratory physics, we explicitly define a **projectile beam** and a **stationary target**

- **In Stars:** Thermonuclear reactions occur within a hot, isotropic plasma. The distinction between "beam" and "target" disappears.
- **The Core Variable:** We care about the number of reactions taking place per unit time per unit volume $\rightarrow$ **Reaction Rates**.

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Nuclear Science

Q-Value

Measures *whether* an isolated reaction occurs and its energy balance.

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Measures *whether* an isolated reaction occurs and its energy balance.

Nuclear Astrophysics Focus

Reaction Rates

Measures *how fast* stars burn energy and synthesize elements.

From Cross Section to Reaction Rates

Stellar reaction rates depend fundamentally on two ingredients

- **Cross section** $\sigma(E)$: microscopic nuclear physics
- **Velocity distribution**: Maxwell–Boltzmann distribution of plasma particles

Stellar Reaction Rates: Identical Particles & Energy Calculus

Particle Identity Correction (The Kronecker Symbol)

To find the volumetric rate r_{01} for any pair of nuclei, we must avoid double-counting interactions if the species are identical (e.g., $p + p$ vs. $p + {}^{12}\text{C}$):

$$r_{01} = \frac{n_0 n_1 \langle \sigma v \rangle_{01}}{1 + \delta_{01}}$$

where δ_{01} is the Kronecker delta: $\delta_{01} = 1$ if species $0 = 1$, and 0 otherwise.

Maxwell-Boltzmann Calculus: Velocity to Energy

Stellar kinetic energy is driven by thermal motion. Using the reduced mass $m_{12} = \frac{m_1 m_2}{m_1 + m_2}$, we transform the relative velocity distribution $P(v)dv$ into an energy distribution $P(E)dE$:

$$E = \frac{1}{2} m_{12} v^2 \implies dE = v m_{12} dv \implies dv = \sqrt{\frac{1}{2m_{12}E}} dE$$

Substituting into $P(v)dv$ yields the energy spectrum form:

Stellar Reaction Rates: Identical Particles & Energy Calculus

Peak Distributions

Most probable velocity: $v_T = \sqrt{\frac{2k_B T}{m_{12}}} \implies$ occurs at energy $E = k_B T$.

Energy distribution peak: Peak of $P(E)$ occurs at $E_T = \frac{1}{2} k_B T$.

The Astrophysical Standard Format

Integrating $\langle \sigma v \rangle = \int_0^\infty v \sigma(E) P(E) dE$ and converting to laboratory units (M_i in amu, E in MeV, σ in barns, and introducing Avogadro's number N_A to yield macroscopic rate units of $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$):

$$N_A \langle \sigma v \rangle_{12} = \frac{3.79 \times 10^{10}}{(T_9)^{3/2}} \sqrt{\frac{M_1 + M_2}{M_1 M_2}} \int_0^\infty E \cdot \sigma(E) \cdot \exp\left(-\frac{11.605 E}{T_9}\right) dE$$

T_9 : Temperature scaled in Giga-Kelvin ($T/10^9$ K).

The exponential constant 11.605 stems directly from $\frac{1}{k_B} \approx 11.605 \text{ GK/MeV}$.

Stellar Reaction Rates: Identical Particles & Energy Calculus

Derivation Proof: Evaluating the Integrand Coefficients

Substitute $v = \sqrt{2E/m_{12}}$ and $P(E)dE$ into the core $\langle\sigma v\rangle = \int_0^\infty v\sigma(E)P(E)dE$ integral:

$$\langle\sigma v\rangle = \int_0^\infty \sqrt{\frac{2E}{m_{12}}} \sigma(E) \left[\frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E} e^{-E/k_B T} \right] dE$$

Combining coefficients ($\sqrt{2} \times 2 = \sqrt{8}$) and terms ($\sqrt{E} \times \sqrt{E} = E$) yields our fundamental analytical step:

$$\langle\sigma v\rangle = \sqrt{\frac{8}{\pi m_{12}}} \frac{1}{(k_B T)^{3/2}} \int_0^\infty E \cdot \sigma(E) \cdot e^{-E/k_B T} dE$$

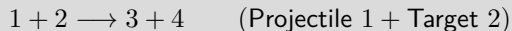
Unit Scale Transformations ($m_{12} \rightarrow \text{amu}$, $T \rightarrow T_9$, N_A)

$$\text{Let } m_{12} = \left(\frac{M_1 M_2}{M_1 + M_2} \right) \cdot m_u = \mu_{\text{amu}} \cdot (1.6605 \times 10^{-24} \text{ g})$$

Stellar Reaction Rates: Fixed Velocity

Reaction Setup in a Stellar Plasma

Consider a general stellar fusion reaction:



Let the cross section be explicitly velocity dependent: $\sigma = \sigma(v)$.

Derivation of Volumetric Rate

The total number of reactions per unit time per unit volume (V) is given by:

$$\begin{aligned} r_{12} &= \frac{N_R}{t \cdot V} \approx \sigma(v) \cdot N_t \cdot \left[\frac{N_b}{V \cdot A \cdot t} \right] && (\text{Substituting } V = A \cdot v \cdot t) \\ &= \sigma(v) \cdot \frac{N_t}{V} \cdot \frac{N_b}{A \cdot t} \\ &= \sigma(v) \cdot \frac{N_t}{V} \cdot \frac{N_b}{V} \cdot v \end{aligned}$$

Stellar Reaction Rates: Velocity Distributions

Accounting for Thermodynamic Velocity Distributions

In a real stellar plasma at temperature T , particles are not monokinetic. Instead, velocities follow a probability distribution $P(v)dv$:

$$P(v)dv = \text{Probability that relative velocity is between } v \text{ and } v + dv$$

where the distribution is normalized: $\int_0^\infty P(v)dv = 1$.

The Integrated Stellar Reaction Rate

Integrating across all possible relative velocities yields the total volumetric rate:

$$r_{12} = n_1 n_2 \int_0^\infty v P(v) \sigma(v) dv$$

We define the **astrophysical reaction rate per particle pair** as the expectation value:

$$\langle \sigma v \rangle_{12} = \int_0^\infty v \sigma(v) P(v) dv$$

Maxwell-Boltzmann Velocity Integration

The Classical Plasma Regime

In standard main-sequence stars like our Sun, the plasma is at high temperature T and low quantum density. Under these conditions, the relative velocities of the interacting nuclei follow a classical

Maxwell-Boltzmann distribution:

$$P(v)dv = \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} \exp \left(-\frac{\mu v^2}{2k_B T} \right) 4\pi v^2 dv$$

where μ is the reduced mass of the two-body system $(1 + 2)$.

Transforming Velocity to Center-of-Mass Energy (E)

Because nuclear cross sections σ are typically measured or calculated as a function of center-of-mass energy ($E = \frac{1}{2}\mu v^2$) rather than relative velocity, we substitute:

$$E = \frac{1}{2}\mu v^2 \implies dE = \mu v dv$$

The Thermal Average Velocity Reaction Rate

Relative Velocity Simplification

The interaction of two moving particles ($\vec{v}_1, \vec{v}_2$) reduces from a two-body problem to a one-body problem using their relative velocity:

$$\vec{v} = \vec{v}_1 - \vec{v}_2$$

Microscopic cross section is velocity dependent: as velocity $v \uparrow$, the de Broglie wavelength $\lambda = \frac{h}{mv} \downarrow$, altering the effective target area.

Thermal Average Notation $\langle \dots \rangle$

The thermal expectation value of any velocity-dependent function $f(v)$ is given by:

$$\langle f(v) \rangle = \int_0^\infty f(v) \cdot P(v) dv$$

Setting $f(v) = v \cdot \sigma(v)$ yields the dynamic factor $\langle \sigma v \rangle$.

Maxwell-Boltzmann Velocity Integration

The Classical Plasma Regime

In stars like our Sun, plasma is at high T and low density. Relative velocities follow the **Maxwell-Boltzmann distribution**:

$$P(v)dv = \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} \exp \left(-\frac{\mu v^2}{2k_B T} \right) 4\pi v^2 dv$$

Astrophysical Reaction Rate Parameter: $\langle \sigma v \rangle$

Converting from relative velocity to center-of-mass energy ($E = \frac{1}{2}\mu v^2$):

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu}} (k_B T)^{-3/2} \int_0^\infty E \sigma(E) \exp \left(-\frac{E}{k_B T} \right) dE$$

Maxwell-Boltzmann Factor: $\exp(-E/k_B T)$ suppresses high-energy particles.

Gamow Factor: $\sigma(E)$ suppresses low-energy interactions due to tunneling.

Comparative Statistics in Stellar Environments

Distribution	Particle Nature	Physical Regime
Maxwell-Boltzmann	Classical, non-interacting	High T , Low Density (Sun)
Fermi-Dirac	Fermions (e^-)	High Density (White Dwarfs)
Bose-Einstein	Bosons (γ , ^4He)	Low T , High Density

Key Case Study

Sun: High temperature and low quantum density $\rightarrow$ **Maxwell-Boltzmann.**

White Dwarfs: Extreme packing ($\rho \uparrow$) $\rightarrow$ **Degenerate Fermi Gas.**

Maxwell-Boltzmann Velocity-to-Energy Conversion

1. Relative Velocity Distribution $P(v)dv$

In classical stellar plasmas, relative velocities follow the Maxwell-Boltzmann profile:

$$P(v)dv = \left(\frac{\mu}{2\pi k_B T} \right)^{3/2} \exp \left(-\frac{\mu v^2}{2k_B T} \right) 4\pi v^2 dv$$

2. Differential Calculus Substitution

To transform this into the center-of-mass energy domain ($E = \frac{1}{2}\mu v^2$), we find the differential relationship:

$$E = \frac{1}{2}\mu v^2 \implies dE = \mu v dv \implies dv = \frac{dE}{\mu v} = \frac{dE}{\sqrt{2\mu E}}$$

Substituting $v^2 = \frac{2E}{\mu}$ and dv into $P(v)dv$ yields the energy distribution spectrum:

$$P(E)dE = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E} \exp \left(-\frac{E}{k_B T} \right) dE$$

Maxwell-Boltzmann Velocity-to-Energy Conversion

3. The Energy-Domain Probability Density $P(E)dE$

$$P(E)dE = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E} \exp\left(-\frac{E}{k_B T}\right) dE$$

4. Evaluating the Thermal Expectation Value $\langle \sigma v \rangle$

We substitute $P(E)dE$ and $v = \sqrt{2E/\mu}$ into the baseline integration setup $\langle \sigma v \rangle = \int_0^\infty v \sigma(E) P(E) dE$:

$$\langle \sigma v \rangle = \int_0^\infty \sqrt{\frac{2E}{\mu}} \sigma(E) \left[\frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E} \exp\left(-\frac{E}{k_B T}\right) \right] dE$$

Simplifying the coefficients ($\sqrt{2} \times 2 = \sqrt{8}$) isolates the fundamental stellar rate formula:

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu}} (k_B T)^{-3/2} \int_0^\infty E \sigma(E) \exp\left(-\frac{E}{k_B T}\right) dE$$

Stellar Reaction Rates: Identical Particles & Energy Calculus

Particle Identity Correction (The Kronecker Symbol)

To find the volumetric rate r_{01} for any pair of nuclei, we must avoid double-counting interactions if the species are identical (e.g., $p + p$ vs. $p + {}^{12}\text{C}$):

$$r_{01} = \frac{n_0 n_1 \langle \sigma v \rangle_{01}}{1 + \delta_{01}}$$

where δ_{01} is the Kronecker delta: $\delta_{01} = 1$ if species $0 = 1$, and 0 otherwise.

Maxwell-Boltzmann Calculus: Velocity to Energy

Stellar kinetic energy is driven by thermal motion. Using the reduced mass $m_{12} = \frac{m_1 m_2}{m_1 + m_2}$, we transform the relative velocity distribution $P(v)dv$ into an energy distribution $P(E)dE$:

$$E = \frac{1}{2} m_{12} v^2 \implies dE = v m_{12} dv \implies dv = \sqrt{\frac{1}{2m_{12}E}} dE$$

Substituting into $P(v)dv$ yields the energy spectrum form:

Stellar Reaction Rates: Identical Particles & Energy Calculus

Peak Distributions

Most probable velocity: $v_T = \sqrt{\frac{2k_B T}{m_{12}}} \implies$ occurs at energy $E = k_B T$.

Energy distribution peak: Peak of $P(E)$ occurs at $E_T = \frac{1}{2} k_B T$.

The Astrophysical Standard Format

Integrating $\langle \sigma v \rangle = \int_0^\infty v \sigma(E) P(E) dE$ and converting to laboratory units (M_i in amu, E in MeV, σ in barns, and introducing Avogadro's number N_A to yield macroscopic rate units of $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$):

$$N_A \langle \sigma v \rangle_{12} = \frac{3.79 \times 10^{10}}{(T_9)^{3/2}} \sqrt{\frac{M_1 + M_2}{M_1 M_2}} \int_0^\infty E \cdot \sigma(E) \cdot \exp\left(-\frac{11.605 E}{T_9}\right) dE$$

T_9 : Temperature scaled in Giga-Kelvin ($T/10^9$ K).

The exponential constant 11.605 stems directly from $\frac{1}{k_B} \approx 11.605 \text{ GK/MeV}$.

Derivation: The Astrophysical Constant 3.79×10^{10}

1. Starting Point: The Analytical Integral

From the Maxwell-Boltzmann energy distribution, the reaction rate per particle pair is:

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu}} (k_B T)^{-3/2} \int_0^\infty E \sigma(E) \exp\left(-\frac{E}{k_B T}\right) dE$$

To find the macroscopic rate $N_A \langle \sigma v \rangle$, we must substitute physical constants:

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1} \text{ (Avogadro's Number)}$$

$$\mu = m_u \cdot \frac{M_1 M_2}{M_1 + M_2} \text{ where } m_u = 1.6605 \times 10^{-24} \text{ g (Atomic Mass Unit)}$$

$$k_B = 8.6173 \times 10^{-5} \text{ MeV/K (Boltzmann Constant in MeV)}$$

Derivation: The Astrophysical Constant 3.79×10^{10}

2. Unit Scaling and Constant Evaluation

We introduce $T_9 = T/10^9$ K and scale the energy-temperature factor:

$$\frac{1}{k_B T} = \frac{1}{(8.6173 \times 10^{-5} \text{ MeV/K})(10^9 \text{ K} \cdot T_9)} = \frac{11.605}{T_9} \text{ MeV}^{-1}$$

Substituting the mass and temperature scales into the coefficient:

$$\text{Coeff} = N_A \cdot \sqrt{\frac{8}{\pi m_u \mu_{\text{amu}}}} \cdot \left(\frac{11.605}{T_9} \right)^{3/2} \times (\text{Unit Conversions})$$

Derivation: The Astrophysical Constant 3.79×10^{10}

3. Final Numerical Result

Grouping the constants (and converting σ from Barns to cm^2 via 10^{-24}):

$$\text{Constant} = \underbrace{6.022 \times 10^{23} \cdot \sqrt{\frac{8}{\pi \cdot 1.66 \times 10^{-24}}} \cdot (11.605)^{3/2}}_{\text{Evaluates to } 3.79 \times 10^{34}} \cdot \underbrace{10^{-24}}_{\text{Barns to cm}^2}$$

$$N_A \langle \sigma v \rangle_{12} = \frac{3.79 \times 10^{10}}{T_9^{3/2}} \sqrt{\frac{M_1 + M_2}{M_1 M_2}} \int_0^\infty \dots dE$$

The units are now strictly in $\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ as required for stellar models.

Reaction Rates and Elemental Abundance

Stellar Nucleosynthesis and Cosmic Abundances

Why are some elements abundant while others are extremely rare? The cosmic evolution and abundance of chemical elements depend directly on the number of nuclear reactions occurring in stellar environments:

High Number of Reactions $\longrightarrow$ High Elemental Abundance

Physical Drivers of Reaction Quantities

The absolute number of nuclear reactions is fundamentally governed by:

Number Density: Total number of available interacting particles.

Cross Section (σ): The intrinsic probability of the nuclear interaction.

Velocity (v): Thermal motion driving particle collisions.

Reaction Rates and Elemental Abundance

Roadmap

In this upcoming framework, we will systematically evaluate:

Neutron-Induced Reactions (e.g., s-process, r-process)

Photon-Induced Reactions (Photodisintegration processes)

Mass Fractions (Tracking chemical composition over time)

Reaction Rates and Elemental Abundance

Mathematical Summary: Maxwell-Boltzmann Review

The velocity distribution translates to energy space as:

$$P(E)dE = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E} \exp\left(-\frac{E}{k_B T}\right) dE$$

The Practical Numerical Expression

Evaluating the constant for laboratory and stellar modeling use:

$$N_A \langle \sigma v \rangle_{12} = \frac{3.79 \times 10^{10}}{(T_9)^{3/2}} \sqrt{\frac{M_1 + M_2}{M_1 M_2}} \int_0^\infty E \cdot \sigma(E) \cdot \exp\left(-\frac{11.605E}{T_9}\right) dE$$

Units Reminder: The output is strictly in $\boxed{\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}}$.

Energy E must be input in **MeV**, and temperature T_9 is in **GK** (10^9 K).

Outline

Thermonuclear Reactions in Stars

Core Nuclear Physics Concepts

Reaction Probabilities and Reaction Rates

Neutron-Induced Reactions

Part III

Neutron-Induced Reactions

Mathematics of Neutron-Induced Reactions

The Absence of the Coulomb Barrier

Since neutrons are neutral ($Z = 0$), the Gamow factor $e^{-2\pi\eta}$ disappears. The cross section is governed strictly by the overlap of the neutron wave with the nucleus.

The $1/v$ Law

At low energies ($E \ll$ resonance energies), the neutron capture cross section follows the $1/v$ relationship:

$$\sigma(v) \propto \frac{1}{v} \implies \sigma(v) = \sigma_0 \frac{v_0}{v}$$

where v_0 is a reference velocity (typically thermal velocity at 2200 m/s).

Implication: As the neutron slows down, its probability of being captured by a nucleus increases dramatically.

Mathematics of Neutron-Induced Reactions

The Thermal Average $\langle \sigma v \rangle$ for Neutrons

Substituting the $1/v$ law into the thermal average integral:

$$\langle \sigma v \rangle = \int_0^\infty v \sigma(v) P(v) dv = \int_0^\infty v \left(\frac{\text{const}}{v} \right) P(v) dv$$

Since the v terms cancel, the integral reduces to the normalization of the probability distribution:

$$\langle \sigma v \rangle = \text{constant}$$

Astrophysical Impact

For $1/v$ absorbers, the reaction rate per particle pair is ****independent of temperature****.

s-process: Slow neutron capture occurs in AGB stars.

r-process: Rapid neutron capture occurs in explosive environments.

This allows neutron captures to proceed efficiently even in "cool" stellar layers where charged-particle 